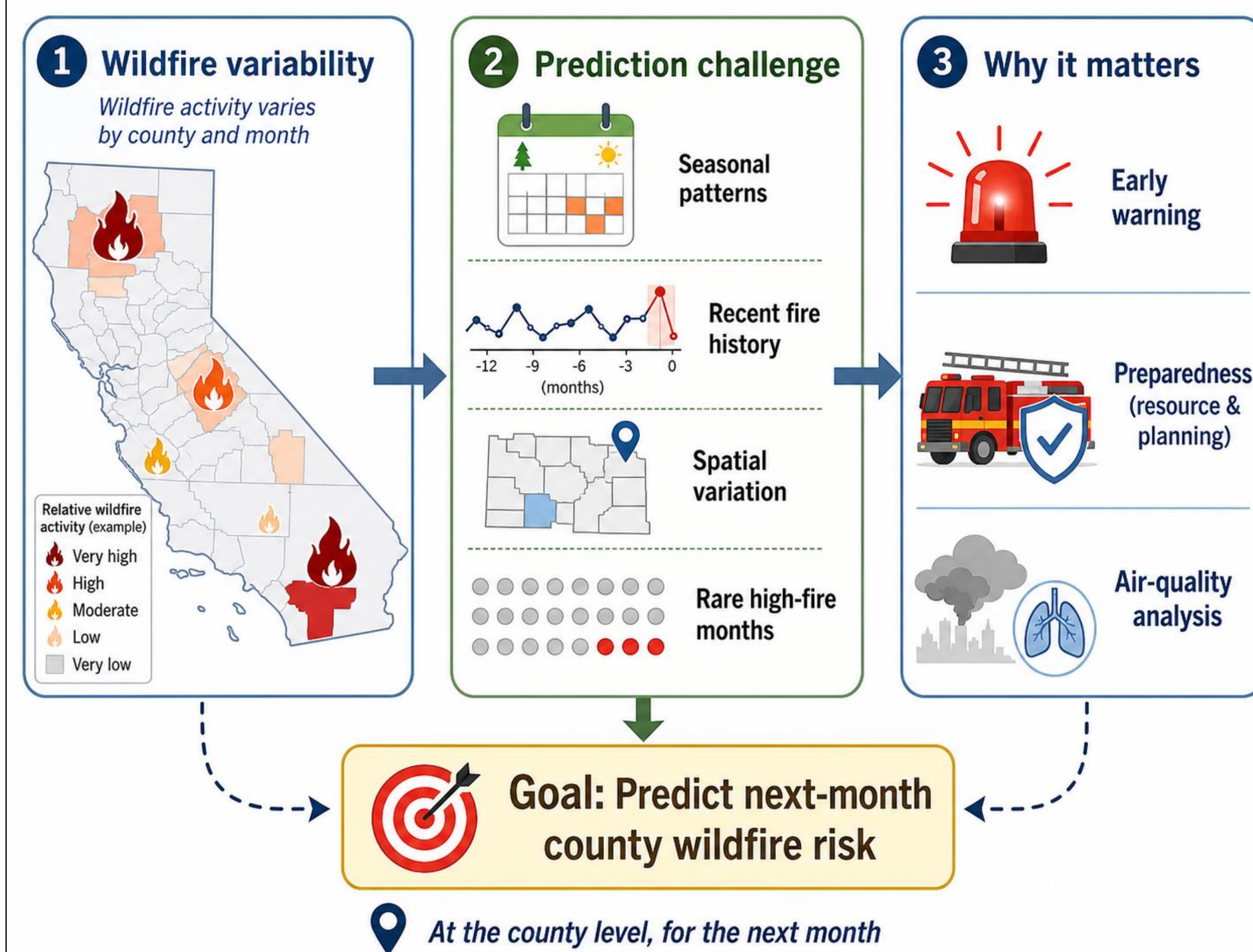


Research Problem

- Increasing wildfire frequency is worsening air quality through elevated PM2.5 (particulate matter) levels, posing serious public health risks.
- Wildfire and PM2.5 patterns vary by county, season, and recent history, with rare high-risk events that require reliable detection.
- This project studies county-level atmospheric risk in California by combining Channel 1 wildfire prediction for high wildfire activity and Channel 2 PM2.5 air-quality analysis for pollution patterns at the county-month level.

Research Question:

How effectively can historical satellite-based thermal anomalies and air quality data predict future wildfires, their impact on PM2.5 levels in California, and the correlation between these phenomena?



Datasets

A **channel** is an independent but connected modeling pipeline for a specific prediction task contributing to the overall analysis

Channel I: Wildfire Prediction

- Data Pipeline:** NASA FIRMS → county-month features

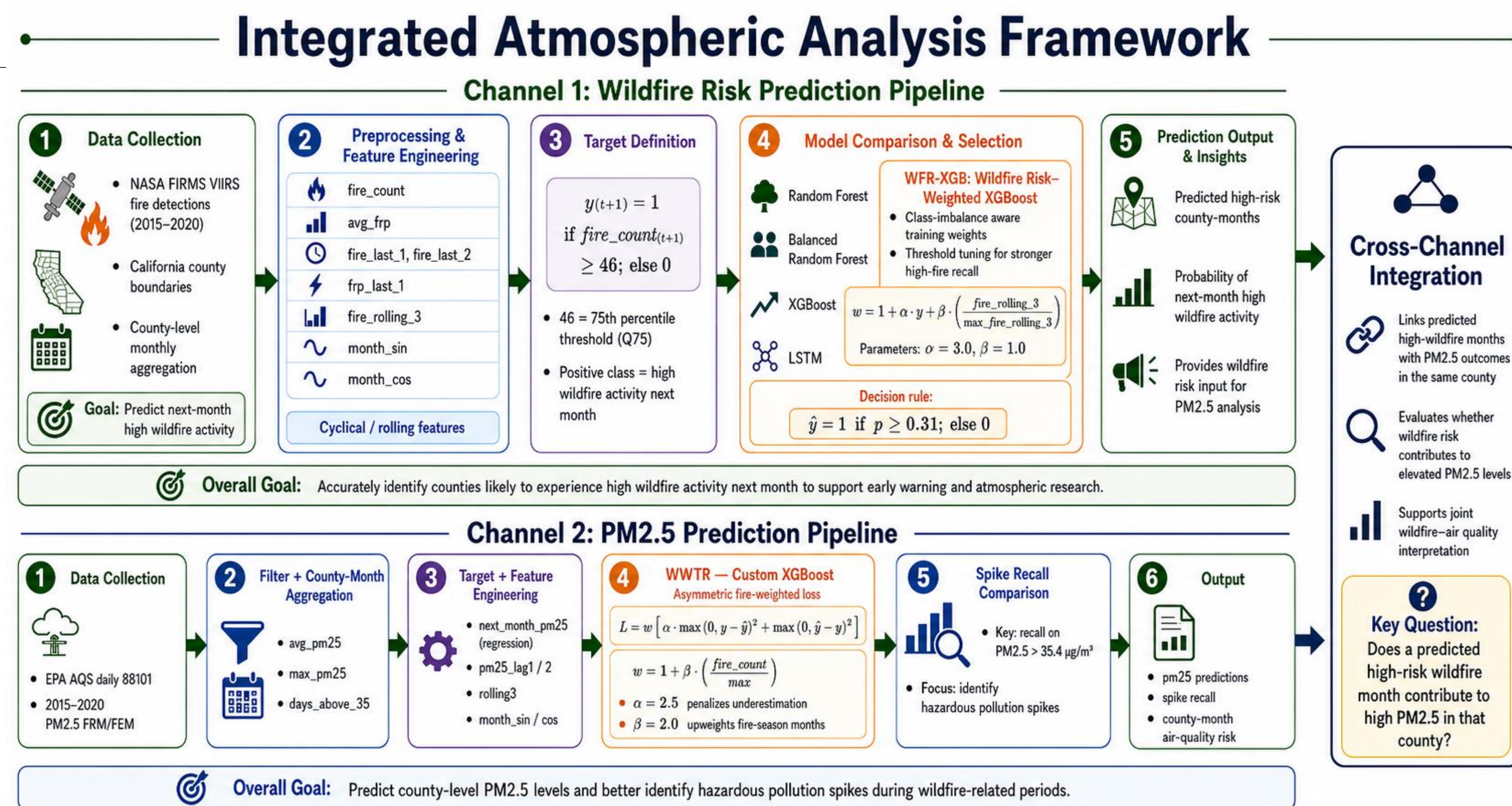
Channel II — PM_{2.5} Air Quality Prediction

- Data Pipeline:** EPA AQS PM 2.5 data (2015–2020), aggregated to county-month (California).

Contribution

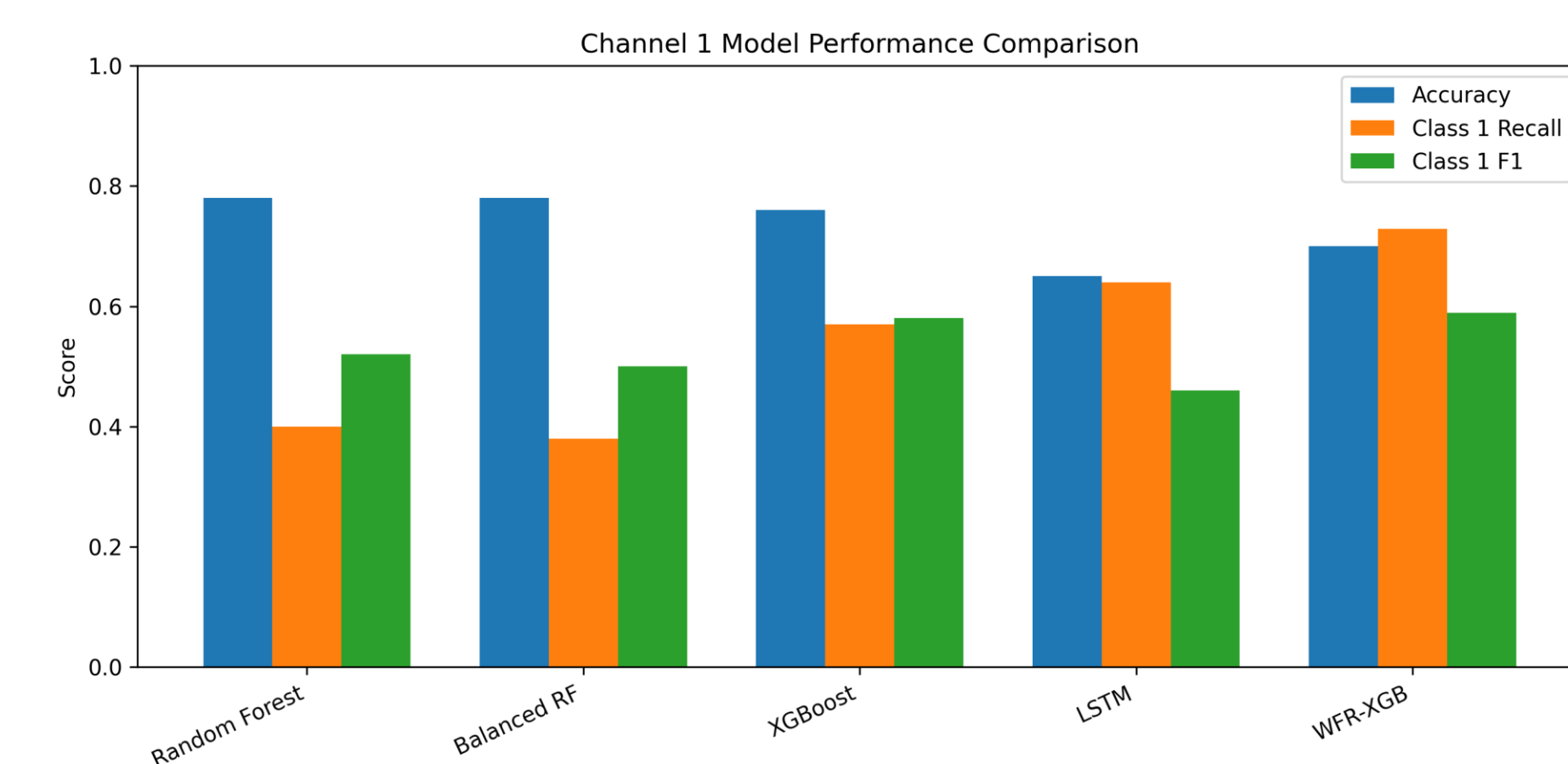
- Dual-task framework:** Designed a two-channel pipeline to jointly model wildfire prediction and PM 2.5 air quality forecasting at the county level.
- Improved wildfire prediction:** Developed **WFR-XGBoost**, increasing recall from **0.57 to 0.73** for high wildfire activity detection.
- Air quality modeling innovation:** Proposed **WWTR**, an asymmetric loss-based model to better capture PM 2.5 spikes during wildfire seasons.
- Cross-domain insight:** Established a predictive link between wildfire activity and PM 2.5 using correlation and VAR-based Granger causality analysis

Proposed Method

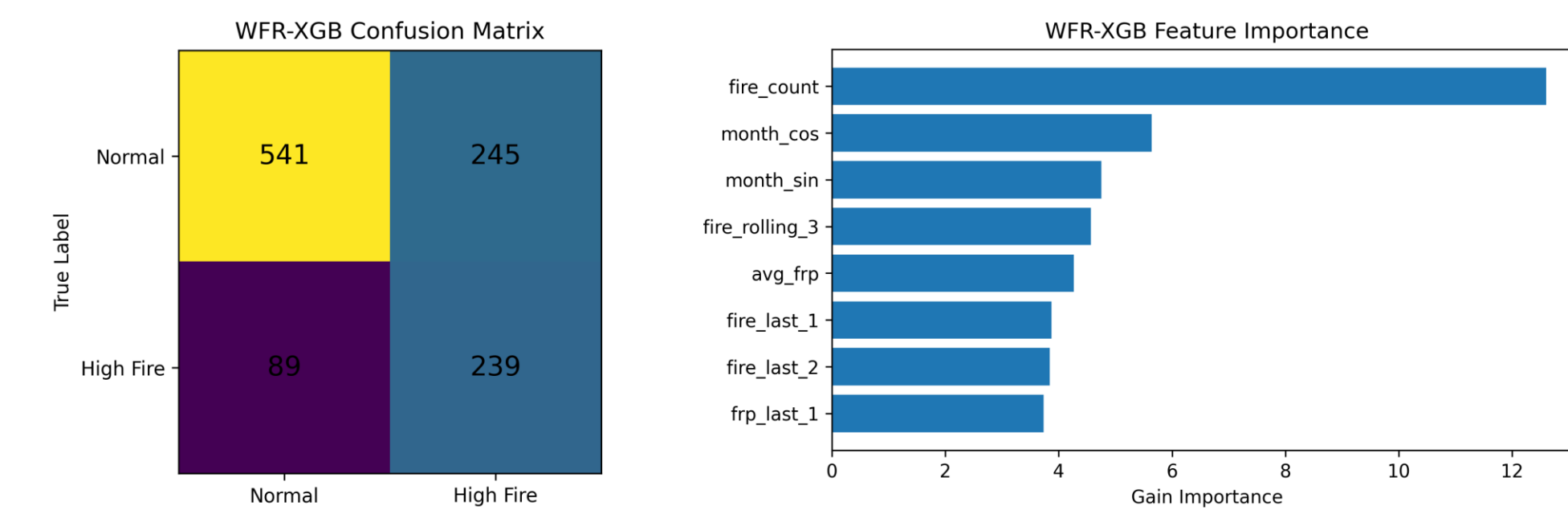


Experimental Results

Channel I:



WFR-XGB improved high-fire recall from 0.57→0.73 compared with standard XGBoost.



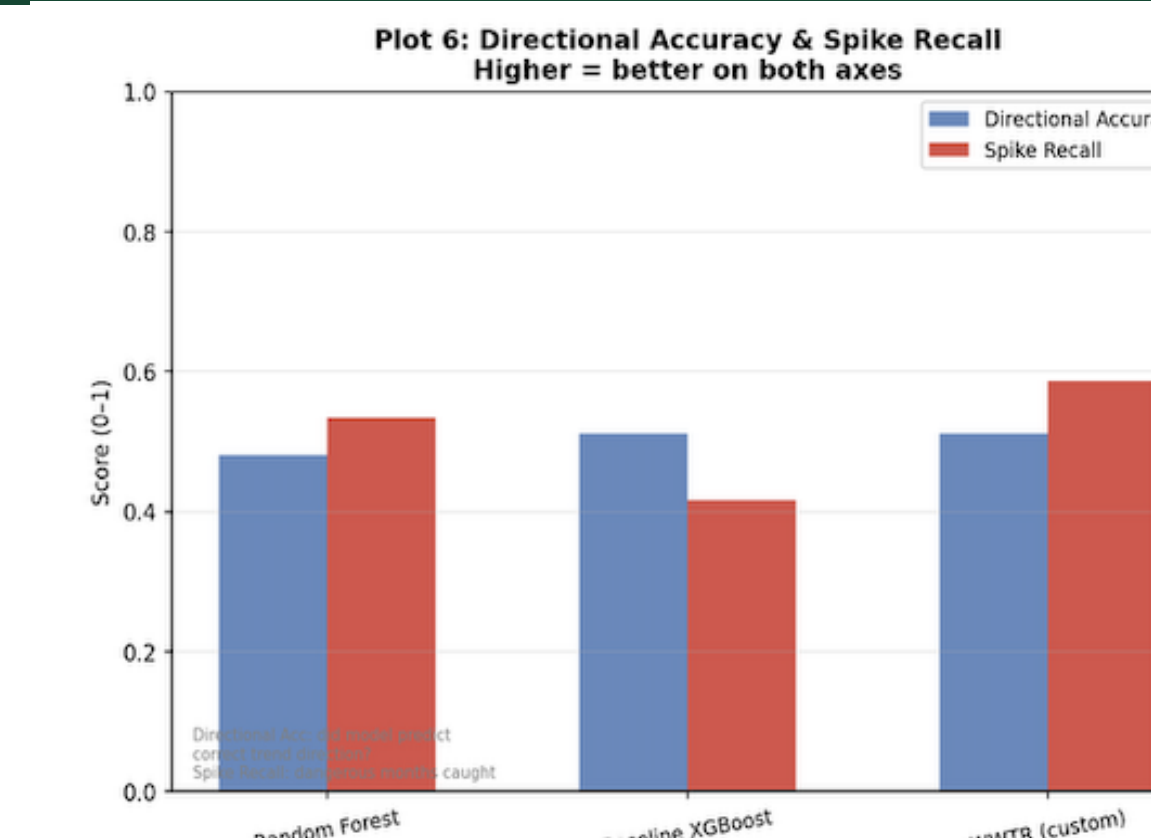
WFR-XGB detected 239 high-fire months and missed 89.

Fire count, seasonality, and recent fire history were the top predictors.

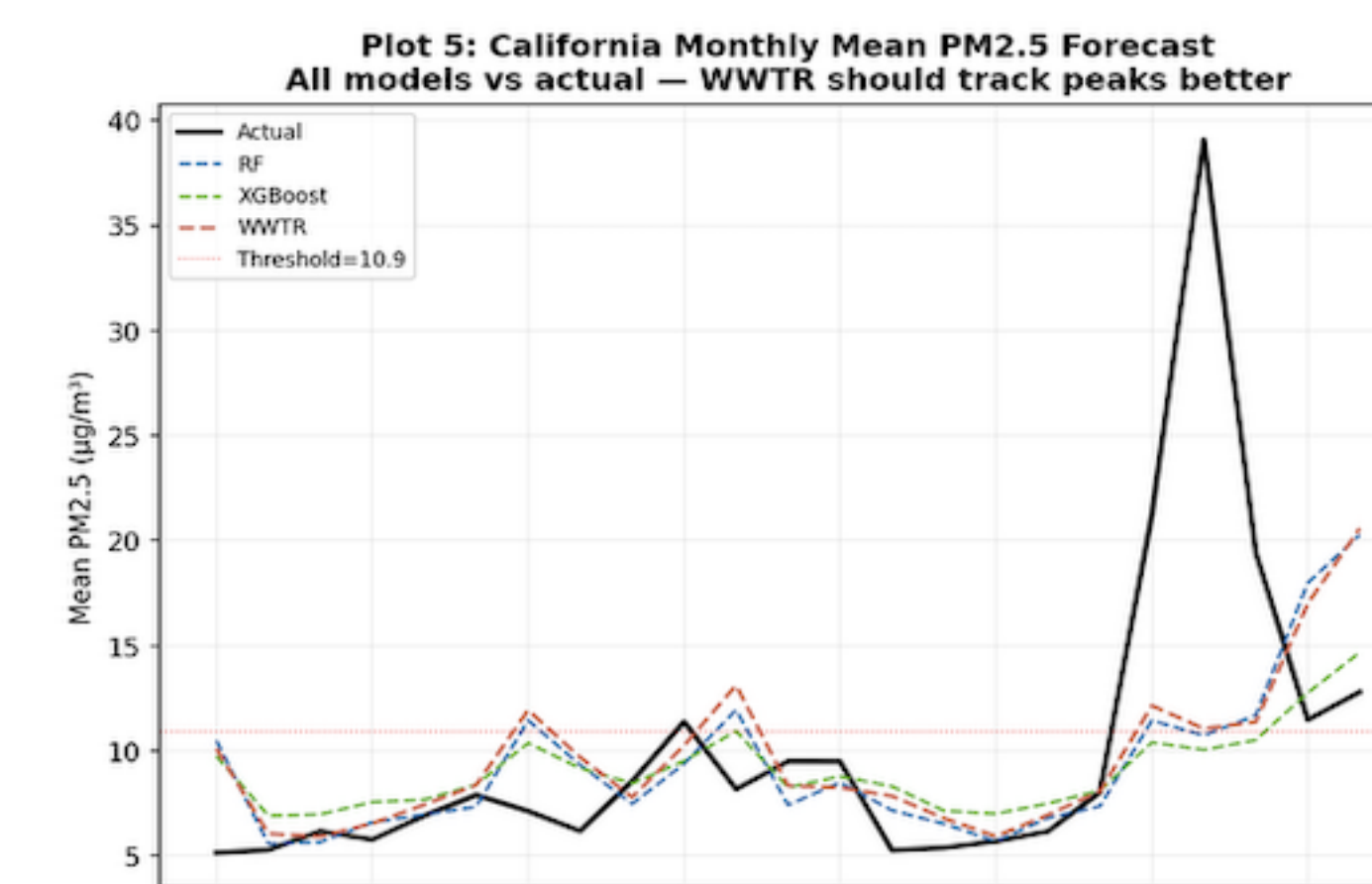
Qualitative Analysis:

Aspect	Insight
Success	More high-fire months detected.
Failure	Some false alarms remain.
Error	Higher recall is prioritized.

Channel II:



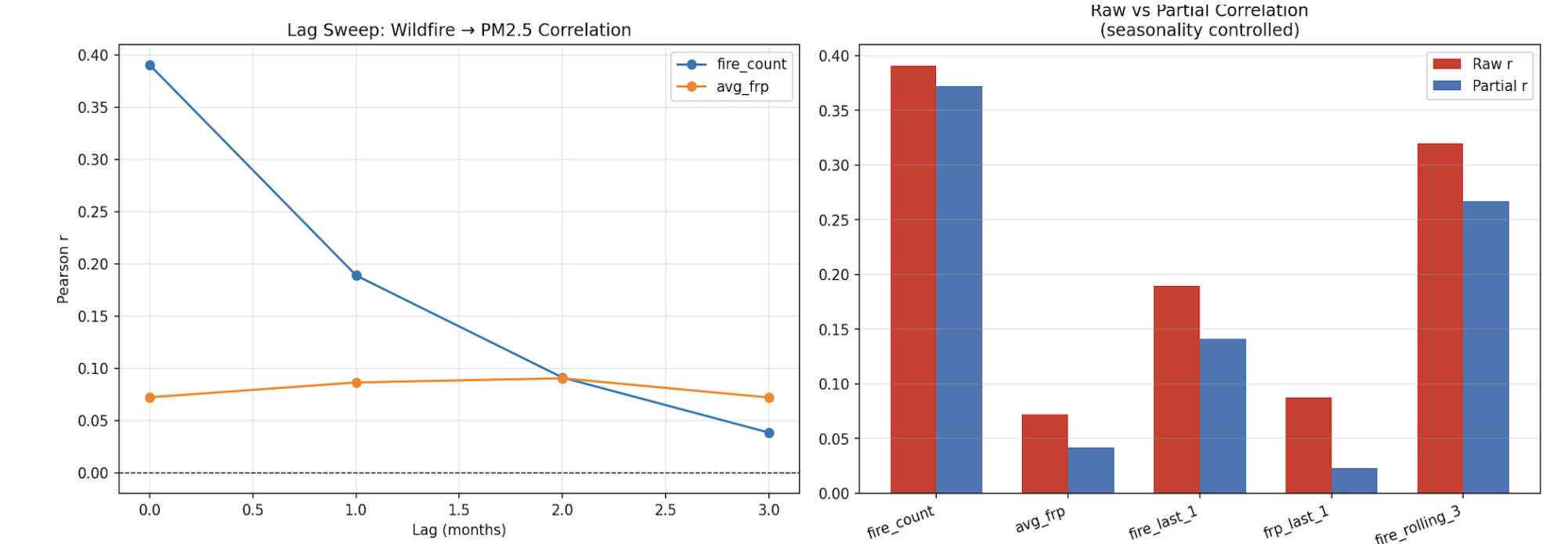
- WWTR achieves highest spike recall:** It captures the most PM2.5 spike events, outperforming other models.
- Balanced performance:** WWTR maintains competitive directional accuracy while improving recall, unlike XGBoost which shows lower spike detection



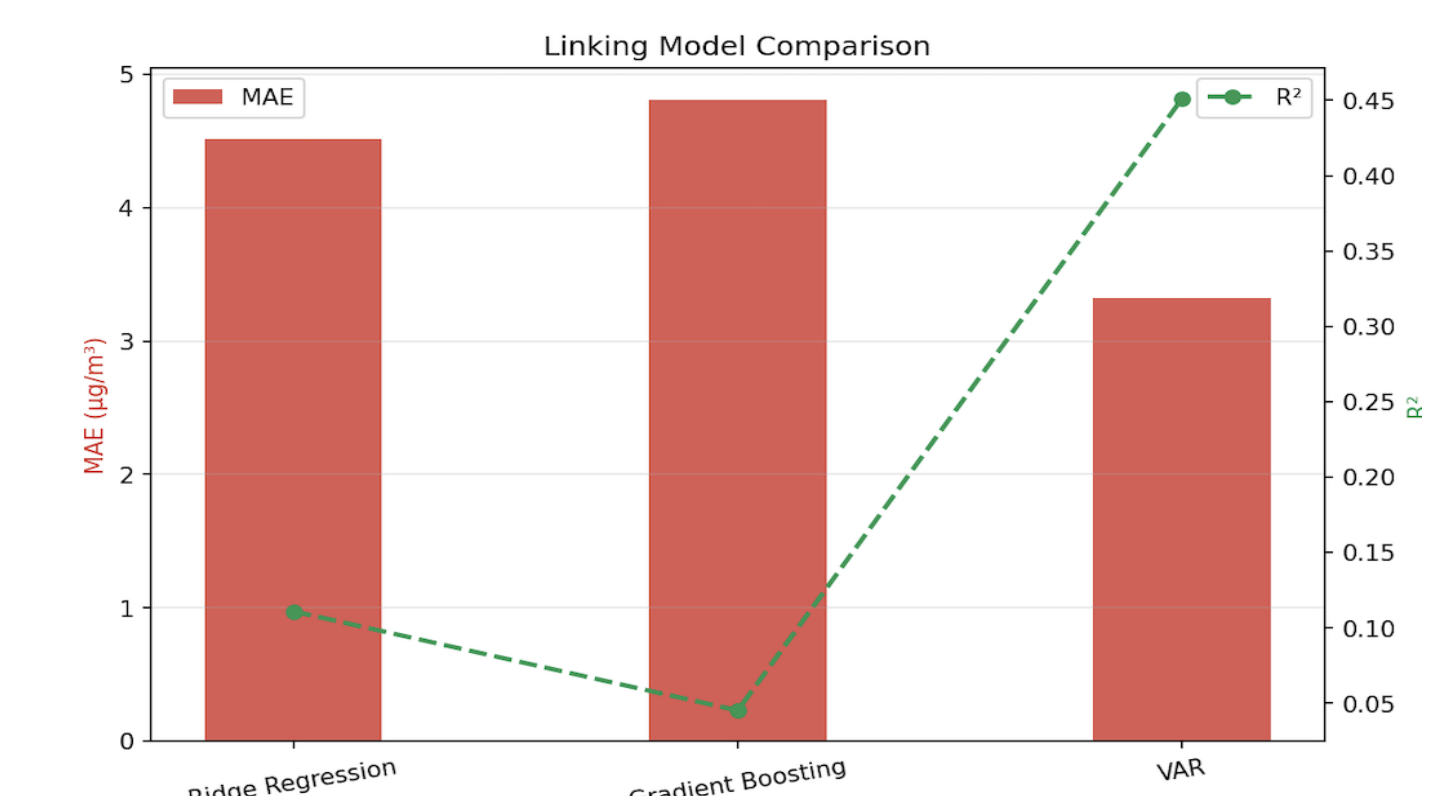
- All models capture general trends:** Predictions follow the overall PM 2.5 pattern, but struggle with extreme spikes.
- WWTR better tracks peaks:** Compared to others, WWTR more closely follows higher PM 2.5 values, improving peak representation.

Experimental Results

Correlation Analysis Result::



- Immediate impact dominates:** Wildfire activity (especially fire count) shows the strongest correlation with PM 2.5 at **0-month lag (~0.39)**, which steadily decreases over time
- Robust but slightly reduced correlation after controls:** key wildfire indicators (e.g., fire count, rolling averages) maintain **moderate positive correlation with PM 2.5**



- VAR performs best:** Lowest MAE (~3.3 $\mu\text{g}/\text{m}^3$) and highest R² (~0.45), indicating better predictive accuracy and fit.
- Gradient Boosting underperforms:** Highest error and lowest R², suggesting weaker modeling of the wildfire–PM 2.5 relationship

Conclusion

- Two-channel predictive framework:** WFR-XGB (wildfire prediction) + WWTR (PM 2.5 forecasting) to model environmental risk at the county level.
- Strong wildfire–air quality linkage:** Linking analysis shows wildfire activity is associated with elevated PM 2.5
- Future Channel (Phase III):** Extend to healthcare by integrating CDC data to predict asthma-related emergency visits and broader public health impacts.

Reference

- [1] NASA FIRMS, VIIRS Active Fire Data, 2015–2020.
- [2] U.S. Census Bureau, TIGER/Line County Shapefiles, 2020.
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- [4] Chen and Guestrin, “XGBoost,” ACM KDD, 2016.
- [5] Hochreiter and Schmidhuber, “Long Short-Term Memory,” Neural Computation, 1997.
- [6] Pedregosa et al., “Scikit-learn,” JMLR, 2011.
- [7] U.S. EPA, AQS PM2.5 Data.